ED Sepsis Management & ICU Admission (simulation)

1. Section I: Scenario Demographics

- **Scenario Title:** Emergency Department Sepsis Management & ICU Admission (Active Siratech HIS Workflow).
- **Target Learning Groups:**
  - Emergency Department (ED) Personnel (Physicians, Nursing, Registration) ○ Health Information Management (HIM) Staff
  - Intensive Care Unit (ICU) Personnel (Faculty, Bed Management) ○ Respiratory Therapy (RT) Staff
  - IT/Health IT Support Staff (Siratech Specialists)
- **Timing Parameters:**
  - **Set-Up:** 30 minutes. ○ **Scenario Execution:** 60 minutes. The scenario duration is specifically designed to simulate the clinical pressure of the **Sepsis 3-hour bundle** and the "Golden Hour" of critical care. ○ **Debriefing:** 30 minutes.

2. Section II: Scenario Developers

- **Key Stakeholders:**
  - Emergency Department Leadership ○ ICU Faculty and Clinical Leads ○ IT/Health IT Support (Siratech Systems Architecture) ○ Nursing Administration & Emergency Respiratory Services ○ Patient Experience and Quality Oversight Committee

3. Section III: Curriculum Integration

- **Educational Goal:** To validate the technical and clinical efficiency of sepsis protocols and the ICU admission workflow within the active Siratech Health Information System (HIS), ensuring institutional standard operating procedures (SOPs) are met under physiological stress.
- **Crisis Resource Management (CRM) Objectives:**
  - **Avoidance of Alert Fatigue:** Appropriate prioritization and response to automated clinical decision support (CDS) triggers within the HIS.
  - **Accuracy of Digital Handover:** Verification of data integrity during the electronic transition of care from the ED dashboard to the ICU admission profile.
  - **Closed-loop Digital Communication:** Confirmation of acknowledgment for critical lab values and STAT medication orders.
- **Operational Objectives:**
  - **HIS Proficiency:** Accurate navigation of the Siratech Sepsis PowerPlan, including SARI score documentation and systemic inflammatory response syndrome (SIRS) criteria entry.

- **Clinical Protocol Adherence:** Execution of the Sepsis Bundle (intravenous fluid resuscitation and antibiotic administration) within the system-mandated timestamps. ○ **Disposition Efficiency:** Coordination of ICU consultation and bed management through the electronic platform to minimize "boarding" time in the ED.

4. Section IV: Scenario Script & Workflow

- **Cast and Equipment:**
  - **Patient:** 75-year-old standardized patient (Gold Stage III COPD). ○ **Confederates:** ED/ICU staff, Respiratory Therapists, and HIM Registrars performing real-world roles. ○ **Required Equipment:** Active Siratech Workstations, IV start kits, pressure bags for fluid boluses, BIPAP/high-flow oxygen systems, and phlebotomy kits for Sepsis-specific lab draws.

Scenario Workflow Table

Step/State	Action / System Trigger	Learner Actions (Siratech Active Protocol)	Auditor Checklist
Step 1: Arrival	Visual Triage at ER entrance.	Perform initial screening; calculate and enter SARI score in HIS.	
Step 2: Registration	System Search / Account Activation.	Conduct electronic MRN lookup; activate patient account for "Emergency" encounter.	
Step 3: Triage	Comprehensive Assessment.	Save vital signs (HR/Temp) to trigger automated Sepsis Alert logic.	Verify "Sepsis Alert" was triggered in HIS when HR >110 and Temp >38.5 were saved.
Step 4: Inroom	Physician Encounter / Order Entry.	Initiate "Sepsis PowerPlan"; enter orders for Lactic Acid, Blood Cultures, and Antibiotics.	
Step 5: Consultation	Clinical Decision.	Initiate electronic request for ICU consultation via the Siratech Consult Module.	Assess time elapsed from "Sepsis Alert" trigger to ICU Consult request.
Step 6: Admission	ICU Acceptance.	Execute electronic bed request; complete admission medication reconciliation.	
Step 7: Transfer	Physical Move / Status Change.	Complete digital handover; update system status to "Transferred."	

5. Section V: Patient Data and Baseline State

- **Clinical Vignette:** A 75-year-old male with a history of COPD presents to the Emergency Department with a 6-hour history of worsening dyspnea and chills. The patient presents with a GCS of 13 (confused to time and place). The clinical presentation is indicative of Septic Shock secondary to pneumonia.
- **Patient Profile:**
  - **Age:** 75 ○ **PMH:** COPD, DM ○ **Allergies:** NKDA.
- **Baseline Vitals:**
  - **O2 Saturation:** 82% on Room Air (RA). ○ **Respiratory Rate (RR):** 28/min. ○ **Heart Rate (HR):** 120 bpm.
  - **Blood Pressure (BP):** 90/50 mmHg. ○ **Temperature:** 39.0°C.

6. Section VI: Supporting Documents & Results

- **Electronic Requisitions:** The following requisitions must be validated as "Active" and "Acknowledged" in Siratech: ○ **Sepsis Labs:** Lactic Acid (Serial), Blood Cultures x2, CBC with Diff, Procalcitonin.
  - **Imaging:** Portable Chest X-ray (Reason: Rule out pneumonia).
  - **Respiratory Support:** BIPAP titration orders and ABG requisitions.
- **System Verification (Consultant Audit):**
  - **Timestamp Audit:** Cross-reference HIS "Order Acknowledged" time for antibiotics with Pharmacy "Dispense" time to identify delivery lags. ○ **Logic Trigger Verification:** Confirm that the Sepsis PowerPlan decision support was triggered based on the specific SIRS criteria (HR >110, Temp >38.5, RR >20) entered during the triage phase.
  - **Data Continuity:** Ensure the patient's MRN matches all digital labels for blood cultures and lab requisitions.

7. Section VII: Debriefing Guide

1. **System Efficiency:** How did the automated Siratech sepsis alerts influence the speed of the 3-hour bundle initiation compared to manual recognition?
2. **Inter-departmental Coordination:** How did the ICU team receive the consultation? Was there any lag between the electronic "Bed Request" and its visibility on the ICU dashboard?
3. **User Experience:** Identify any software friction points where the system navigation hindered clinical focus during the patient's resuscitation.
4. **Follow-up Coordination:** Based on the "Patient Experience" oversight, how will the digital records facilitate the transition to OPD follow-up post-discharge?